ARTIFICIAL INTELLIGENCE ALL YEARS

===
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
===

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 1
QUESTION: What is NLP?
OPTIONS: (A) Natural Language Processing (B) Neural Language Programming (C) Non-linear Programming (D) Numerical Logic Processing
ANSWER: (A) Natural Language Processing

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 1
QUESTION: Mention any two commonly used applications of NLP.
OPTIONS: (A) Sentiment Analysis and Machine Translation (B) Image Recognition and Speech Synthesis (C) Database Management and Web Scraping (D) Network Security and Cryptography
ANSWER: (A) Sentiment Analysis and Machine Translation

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 1
QUESTION: Name any two currently popular virtual assistants.
OPTIONS: (A) Siri and Alexa (B) Google Maps and YouTube (C) Windows and Linux (D) Photoshop and Illustrator
ANSWER: (A) Siri and Alexa

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 1

QUESTION: Name the process of dividing whole corpus into sentences.
OPTIONS: (A) Sentence Segmentation (B) Tokenization (C) Stemming (D) Lemmatization
ANSWER: (A) Sentence Segmentation

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 1
QUESTION: With reference to evaluation process of understanding the reliability of any AI model, define the term True Positive.
OPTIONS: (A) Correctly predicted positive cases (B) Incorrectly predicted positive cases (C) Correctly predicted negative cases (D) Incorrectly predicted negative cases
ANSWER: (A) Correctly predicted positive cases

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 1
QUESTION: What is F1 score?
OPTIONS: (A) Harmonic mean of Precision and Recall (B) Arithmetic mean of Precision and Recall (C) Geometric mean of Precision and Recall (D) Sum of Precision and Recall
ANSWER: (A) Harmonic mean of Precision and Recall

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: What is the purpose of Evaluation stage of AI project cycle? Discuss briefly.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: The Evaluation stage tests the model's performance using metrics like accuracy, precision, recall, and F1-score to ensure it generalises well to new data.

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2

QUESTION: What is Tokenization? Count how many tokens are present in the following statement: "I find that the harder I work, the more luck I seem to have."
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Tokenization is the process of breaking text into smaller units called tokens. The given sentence has 15 tokens.

YEAR: 2022
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: With reference to evaluation stage of AI project cycle, explain the term Accuracy. Also give the formula to calculate it.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Accuracy is the proportion of correct predictions out of total predictions. Formula: (TP+TN)/(TP+TN+FP+FN).

YEAR: 2023
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Identify the incorrect statements from the following: (i) AI models can be broadly categorized into four domains. (ii) Data sciences is one of the domain of AI model. (iii) Price comparison websites are examples of data science. (iv) The information extracted through data science can be used to make decision about it.
OPTIONS: (A) Only (iv) (B) (iii) and (iv) (C) Only (i) (D) (ii) and (iii)
ANSWER: (C) Only (i)

YEAR: 2023
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: During Data Acquisition, feeding previous data into the machine is called:
OPTIONS: (A) Training Data (B) Predicting Data (C) Testing Data (D) Evaluating Data
ANSWER: (A) Training Data

YEAR: 2023

CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is defined as the measure of balance between precision and recall?
OPTIONS: (A) Accuracy (B) F1 Score (C) Reliability (D) Punctuality
ANSWER: (B) F1 Score

YEAR: 2023
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: 4Ws Problem Canvas is a part of:
OPTIONS: (A) Problem Scoping (B) Data Acquisition (C) Modelling (D) Evaluation
ANSWER: (A) Problem Scoping

YEAR: 2023
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a feature of document classification?
OPTIONS: (A) Helps in classifying the type and genre of a document. (B) Helps in creating a document. (C) Helps to display important information of a corpus. (D) Helps in including the necessary words in the text body.
ANSWER: (A) Helps in classifying the type and genre of a document.

YEAR: 2023
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: With reference to AI domain, expand the term CV.
OPTIONS: (A) Computer Vision (B) Computer Version (C) Control Variable (D) Central View
ANSWER: (A) Computer Vision

YEAR: 2023
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI

ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Explain any one example of AI bias.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: AI bias example: Virtual assistants often default to female voices, reflecting gender bias.

YEAR: 2023
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Define Chatbot. What are its types?
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: A chatbot is an AI program that simulates human conversation. Types: Script-bot and Smart-bot.

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Statement 1: In "When" block of 4Ws canvas we find the stakeholders. Statement 2: Stakeholders are the people who face a particular problem and would be benefitted with the solution.
OPTIONS: (A) Both Statement 1 and Statement 2 are correct. (B) Both Statement 1 and Statement 2 are incorrect. (C) Statement 1 is correct but Statement 2 is incorrect. (D) Statement 2 is correct but Statement 1 is incorrect.
ANSWER: (D) Statement 2 is correct but Statement 1 is incorrect.

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Whenever we want an AI project to be able to predict an output, we need to
OPTIONS: (A) first test it using the data. (B) first train it using the data. (C) Both (A) and (B). (D) Neither (A) nor (B).
ANSWER: (B) first train it using the data.

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the AI project cycle, which of the following represents the correct order of steps?
OPTIONS: (A) Data Exploration, Problem Scoping, Modelling, Evaluation, Data Acquisition. (B) Problem Scoping, Data Acquisition, Data Exploration, Modelling, Evaluation. (C) Modelling, Data Acquisition, Evaluation, Problem Scoping, Data Exploration. (D) Data Acquisition, Data Exploration, Problem Scoping, Modelling, Evaluation.
ANSWER: (B) Problem Scoping, Data Acquisition, Data Exploration, Modelling, Evaluation.

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which stage of the AI Project Cycle involves testing the model on newly fetched data?
OPTIONS: (A) Data Exploration (B) Modelling (C) Evaluation (D) Deployment
ANSWER: (C) Evaluation

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): Bioethics is an example of a Value-based Framework for AI. Reason (R): Bioethics deals with ethical issues related to health, medicine, and biological sciences.
OPTIONS: (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.
ANSWER: (D) (A) is false, but (R) is true.

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Ethical frameworks are primarily designed to:

OPTIONS: (A) Increase the efficiency of AI algorithms. (B) Ensure that choices made do not cause unintended harm. (C) Reduce the cost of AI development. (D) Speed up the AI project cycle.
ANSWER: (B) Ensure that choices made do not cause unintended harm.

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: As AI is essentially being used as a decision making / influencing tool, we need to ensure that AI makes morally acceptable recommendations. Which of the following is a key factor that can knowingly or unknowingly influences our decision-making while designing an AI model?
OPTIONS: (A) Intuition and Values (B) Algorithm efficiency (C) Data storage capacity (D) Processing speed
ANSWER: (A) Intuition and Values

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The 4 W's Problem Canvas helps in identifying the key elements related to the given problem. Which of the following is NOT one of the blocks of the Problem Canvas?
OPTIONS: (A) When (B) Where (C) What (D) Why
ANSWER: (D) Why

YEAR: 2024
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: What role does data play in AI based applications? Name any two sources of online data collection for building any AI based application.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Data is the foundation of AI applications; it is used to train, validate, and test AI models. Two online data sources: government open data portals and social media APIs.

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Statement 1: In "When" block of 4Ws canvas we find the stakeholders. Statement 2: Stakeholders are the people who face a particular problem and would be benefitted with the solution.
OPTIONS: (A) Both Statement 1 and Statement 2 are correct. (B) Both Statement 1 and Statement 2 are incorrect. (C) Statement 1 is correct but Statement 2 is incorrect. (D) Statement 2 is correct but Statement 1 is incorrect.
ANSWER: (D) Statement 2 is correct but Statement 1 is incorrect.

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Whenever we want an AI project to be able to predict an output, we need to
OPTIONS: (A) first test it using the data. (B) first train it using the data. (C) Both (A) and (B). (D) Neither (A) nor (B).
ANSWER: (B) first train it using the data.

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: 4Ws Problem Canvas is a part of:
OPTIONS: (A) Problem Scoping (B) Data Acquisition (C) Modelling (D) Evaluation
ANSWER: (A) Problem Scoping

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the AI project cycle, which of the following represents the correct order of steps?
OPTIONS: (A) Data Exploration, Problem Scoping, Modelling, Evaluation, Data Acquisition. (B) Problem Scoping, Data Acquisition, Data Exploration, Modelling, Evaluation. (C) Modelling, Data Acquisition, Evaluation, Problem Scoping, Data Exploration. (D) Data Acquisition, Data Exploration, Problem Scoping, Modelling, Evaluation.

ANSWER: (B) Problem Scoping, Data Acquisition, Data Exploration, Modelling, Evaluation.

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which stage of the AI Project Cycle involves testing the model on newly fetched data?
OPTIONS: (A) Data Exploration (B) Modelling (C) Evaluation (D) Deployment
ANSWER: (C) Evaluation

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Ethical frameworks are primarily designed to:
OPTIONS: (A) Increase the efficiency of AI algorithms. (B) Ensure that choices made do not cause unintended harm. (C) Reduce the cost of AI development. (D) Speed up the AI project cycle.
ANSWER: (B) Ensure that choices made do not cause unintended harm.

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: As AI is essentially being used as a decision making / influencing tool, we need to ensure that AI makes morally acceptable recommendations. Which of the following is a key factor that can knowingly or unknowingly influences our decision-making while designing an AI model?
OPTIONS: (A) Intuition and Values (B) Algorithm efficiency (C) Data storage capacity (D) Processing speed
ANSWER: (A) Intuition and Values

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: Assertion (A): Bioethics is an example of a Value-based Framework for AI. Reason (R): Bioethics deals with ethical issues related to health, medicine, and biological sciences.
OPTIONS: (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.
ANSWER: (D) (A) is false, but (R) is true.

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Define the following with respect to AI project cycle: (i) Data Exploration (ii) Data Features
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (i) Data Exploration is the 3rd step of the AI Project Cycle to analyse the data and discover hidden patterns, trends and relationships. (ii) Data Features are individual measurable properties or characteristics of a dataset used to analyze and build models.

YEAR: 2025
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 4
QUESTION: What do you understand by AI Bias and AI Access? Give one example of each to support your answer.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: AI Bias refers to the presence of discrimination in AI systems that leads to favor certain groups over others. Example: Virtual assistants predominantly have female voices. AI Access refers to the availability to use AI technologies. Example: AI-powered mobile health apps like Babylon Health.

YEAR: 2026
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Anita and Surjit are creating an AI application that will classify different types of fruits. The Computer Vision task that will identify the type of fruit and assign a label to it is called:

OPTIONS: (A) Segmentation (B) Classification (C) Classification + Localization (D) Object Detection
ANSWER: (B) Classification

YEAR: 2026
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following contributes to the efficiency of an AI project?
OPTIONS: (A) High Model Complexity (B) Relevant and Authentic Training Data (C) Minimal Preprocessing (D) Limited Hardware Resources
ANSWER: (B) Relevant and Authentic Training Data

YEAR: 2026
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the AI project cycle, which of the following represents the correct order of steps?
OPTIONS: (A) Data Exploration, Problem Scoping, Modelling, Evaluation, Data Acquisition. (B) Problem Scoping, Data Acquisition, Data Exploration, Modelling, Evaluation. (C) Modelling, Data Acquisition, Evaluation, Problem Scoping, Data Exploration. (D) Data Acquisition, Data Exploration, Problem Scoping, Modelling, Evaluation.
ANSWER: (B) Problem Scoping, Data Acquisition, Data Exploration, Modelling, Evaluation.

YEAR: 2026
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which stage of the AI Project Cycle involves testing the model on newly fetched data?
OPTIONS: (A) Data Exploration (B) Modelling (C) Evaluation (D) Deployment
ANSWER: (C) Evaluation

YEAR: 2026
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: Ethical frameworks are primarily designed to:
OPTIONS: (A) Increase the efficiency of AI algorithms. (B) Ensure that choices made do not cause unintended harm. (C) Reduce the cost of AI development. (D) Speed up the AI project cycle.
ANSWER: (B) Ensure that choices made do not cause unintended harm.

YEAR: 2026
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: As AI is essentially being used as a decision making / influencing tool, we need to ensure that AI makes morally acceptable recommendations. Which of the following is a key factor that can knowingly or unknowingly influences our decision-making while designing an AI model?
OPTIONS: (A) Intuition and Values (B) Algorithm efficiency (C) Data storage capacity (D) Processing speed
ANSWER: (A) Intuition and Values

YEAR: 2026
CHAPTER: 1 - Revisiting AI Project Cycle & Ethical Frameworks for AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): Bioethics is an example of a Value-based Framework for AI. Reason (R): Bioethics deals with ethical issues related to health, medicine, and biological sciences.
OPTIONS: (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.
ANSWER: (D) (A) is false, but (R) is true.

==
CHAPTER: 2 - Advanced Concepts of Modeling in AI
==

YEAR: 2022
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 1

QUESTION: Regression is one of the type of supervised learning model, where data is classified according to labels and data need not to be continuous. (True / False)
OPTIONS: (A) True (B) False
ANSWER: (B) False (Regression predicts continuous values, not classification)

YEAR: 2023
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is the correct feature of Neural network?
OPTIONS: (A) It can improve the efficiency of two models. (B) It is useful with small dataset. (C) They are modelled on human brains and nervous system. (D) They need human intervention.
ANSWER: (C) They are modelled on human brains and nervous system.

YEAR: 2023
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In this learning model, the data set which is fed to the machine is labelled. Name the model.
OPTIONS: (A) Supervised Learning (B) Unsupervised Learning (C) Reinforcement Learning (D) Semi-supervised Learning
ANSWER: (A) Supervised Learning

YEAR: 2023
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: It refers to the unsupervised learning algorithm which can cluster the unknown data according to the patterns or trends identified out of it.
OPTIONS: (A) Regression (B) Classification (C) Clustering (D) Dimensionality reduction
ANSWER: (C) Clustering

YEAR: 2024
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: Which learning approach would be most suitable for training an AI model to park the car correctly?
OPTIONS: (A) Supervised Learning (B) Unsupervised Learning (C) Transfer Learning (D) Reinforcement Learning
ANSWER: (D) Reinforcement Learning

YEAR: 2024
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which scenario best represents a regression problem?
OPTIONS: (A) Identifying whether an email is spam. (B) Grouping customers by behaviour. (C) Predicting tomorrow's temperature. (D) Recognizing faces in photos.
ANSWER: (C) Predicting tomorrow's temperature.

YEAR: 2024
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: An e-commerce platform analyzes customer purchase patterns to recommend "Customers who bought product X also bought product Y." This uses:
OPTIONS: (A) Classification model (B) Regression model (C) Association model (D) Clustering model
ANSWER: (C) Association model

YEAR: 2024
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: When a machine possesses the ability to mimic human traits, i.e., make decisions, predict the future, learn, and improve on its own, it is said to have:
OPTIONS: (A) Computational Skills (B) Learning Capability (C) Artificial Intelligence (D) Cognitive Processing
ANSWER: (C) Artificial Intelligence

YEAR: 2024
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Give any two characteristics of a Classification Model.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (1) It is a type of Supervised Learning model where data is classified according to labels. (2) The model works on a discrete data set which means the data need not be continuous.

YEAR: 2024
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Give two differences between Supervised and Unsupervised learning.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Supervised Learning uses labelled data and predicts outcomes, while Unsupervised Learning uses un-labelled data and discovers hidden patterns. Supervised Learning is less complex and uses user supervision, while Unsupervised Learning is more complex without user supervision.

YEAR: 2024
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 4
QUESTION: What are Neural networks? Briefly explain all the layers of a neural network.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Neural networks are systems of machine learning algorithms organised into three layers: input layer (acquires data), hidden layers (process data using algorithms), and output layer (gives final result).

YEAR: 2025
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which learning approach would be most suitable for training an AI model to park the car correctly?

OPTIONS: (A) Supervised Learning (B) Unsupervised Learning (C) Transfer Learning (D) Reinforcement Learning
ANSWER: (D) Reinforcement Learning

YEAR: 2025
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which scenario best represents a regression problem?
OPTIONS: (A) Identifying whether an email is spam. (B) Grouping customers by behaviour. (C) Predicting tomorrow's temperature. (D) Recognizing faces in photos.
ANSWER: (C) Predicting tomorrow's temperature.

YEAR: 2025
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: An e-commerce platform analyzes customer purchase patterns to recommend "Customers who bought product X also bought product Y." This uses:
OPTIONS: (A) Classification model (B) Regression model (C) Association model (D) Clustering model
ANSWER: (C) Association model

YEAR: 2025
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Give any two characteristics of a Classification Model.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (1) It is a type of Supervised Learning model where data is classified according to labels. (2) The model works on a discrete data set which means the data need not be continuous.

YEAR: 2026
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: Which learning approach would be most suitable for training an AI model to park the car correctly?
OPTIONS: (A) Supervised Learning (B) Unsupervised Learning (C) Transfer Learning (D) Reinforcement Learning
ANSWER: (D) Reinforcement Learning

YEAR: 2026
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which scenario best represents a regression problem?
OPTIONS: (A) Identifying whether an email is spam. (B) Grouping customers by behaviour. (C) Predicting tomorrow's temperature. (D) Recognizing faces in photos.
ANSWER: (C) Predicting tomorrow's temperature.

YEAR: 2026
CHAPTER: 2 - Advanced Concepts of Modeling in AI
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: An e-commerce platform analyzes customer purchase patterns to recommend "Customers who bought product X also bought product Y." This uses:
OPTIONS: (A) Classification model (B) Regression model (C) Association model (D) Clustering model
ANSWER: (C) Association model

==
CHAPTER: 3 - Evaluating Models
==

YEAR: 2022
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 4
QUESTION: Traffic Jams have become a common part of our lives now-a-days. Living in an urban area means you have to face traffic each and every time you get out on the road. Mostly, school students opt for buses to go to school. Many times the bus gets late due to such jams and students are not able to reach their school on time. Thus, an AI model is created to predict

explicitly if there would be a traffic jam on their way to school or not. The confusion matrix for the same is: (matrix shown) Explain the process of calculating F1 score for the given problem.
OPTIONS: This is a numerical question based on a confusion matrix. Not converted to MCQ.
ANSWER: F1 score = 2 * (Precision * Recall) / (Precision + Recall). Precision = TP/(TP+FP), Recall = TP/(TP+FN). Compute using given matrix values.

YEAR: 2023
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: While evaluating a model's performance, recall parameter considers (i) False positive (ii) True positive (iii) False negative (iv) True negative. Choose the correct option:
OPTIONS: (A) only (i) (B) (ii) and (iii) (C) (iii) and (iv) (D) (i) and (iv)
ANSWER: (B) (ii) and (iii) [Recall = TP/(TP+FN)]

YEAR: 2023
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: When the prediction matches the reality, the condition is termed as:
OPTIONS: (A) True Positive (B) True Negative (C) True Positive and True Negative (D) False Positive and False Negative
ANSWER: (C) True Positive and True Negative

YEAR: 2023
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Define Confusion Matrix.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: A confusion matrix is a table used to evaluate the performance of a classification model, showing True Positives, True Negatives, False Positives, and False Negatives.

YEAR: 2024
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: Statement 1: Confusion matrix is an evaluation metric. Statement 2: Confusion Matrix is a record which helps in evaluation.
OPTIONS: (A) Both Statement 1 and Statement 2 are correct. (B) Both Statement 1 and Statement 2 are incorrect. (C) Statement 1 is correct and Statement 2 is incorrect. (D) Statement 2 is correct and Statement 1 is incorrect.
ANSWER: (A) Both Statement 1 and Statement 2 are correct.

YEAR: 2024
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: It is one of the parameters for evaluating a model's performance and is defined as the percentage of true positive cases versus all the cases where the prediction is true. Which of the following evaluation parameter is this?
OPTIONS: (A) Precision (B) Recall (C) F1 score (D) Accuracy
ANSWER: (A) Precision

YEAR: 2024
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Precision is defined as:
OPTIONS: (A) The ratio of correctly predicted positive observations to total observations. (B) The ratio of correctly predicted positive observations to total predicted positive observations. (C) The ratio of correctly predicted negative observations to total observations. (D) The harmonic mean of true positives and true negatives.
ANSWER: (B) The ratio of correctly predicted positive observations to total predicted positive observations.

YEAR: 2024
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Statement 1: To evaluate a model's performance, we need either precision or recall. Statement 2: When the value of both Precision and Recall is 1, the F1 score is 0.

OPTIONS: (A) Both statement 1 and statement 2 are correct. (B) Both statement 1 and statement 2 are incorrect. (C) Statement 1 is correct, but statement 2 is incorrect. (D) Statement 1 is incorrect, but statement 2 is correct.
ANSWER: (C) Statement 1 is correct, but statement 2 is incorrect.

YEAR: 2024
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: What is the primary need for evaluating an AI model's performance in the AI Model Development process?
OPTIONS: (A) To increase the complexity of the model. (B) To visualize the data. (C) To assess how well the chosen model will work in future. (D) To reduce the amount of data used for training.
ANSWER: (C) To assess how well the chosen model will work in future.

YEAR: 2024
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: What do you mean by Evaluation of an AI model? Also explain the concept of overfitting with respect to AI model Evaluation.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Evaluation is the process of testing an AI model's performance. Overfitting occurs when a model memorizes training data rather than learning patterns, performing well on training data but poorly on new data.

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Statement 1: Overfitting is not recommended for evaluation of a model. Statement 2: This is because the model will simply remember the whole training set, and will therefore always predict the correct label for any point in the training set.
OPTIONS: (A) Both Statement 1 and Statement 2 are correct. (B) Both Statement 1 and Statement 2 are incorrect. (C) Statement 1 is correct but Statement 2 is incorrect. (D) Statement 2 is correct but Statement 1 is incorrect.
ANSWER: (A) Both Statement 1 and Statement 2 are correct.

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: It is one of the parameters for evaluating a model's performance and is defined as the percentage of true positive cases versus all the cases where the prediction is true. Which of the following evaluation parameter is this?
OPTIONS: (A) Precision (B) Recall (C) F1 Score (D) Accuracy
ANSWER: (A) Precision

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Precision is defined as:
OPTIONS: (A) The ratio of correctly predicted positive observations to total observations. (B) The ratio of correctly predicted positive observations to total predicted positive observations. (C) The ratio of correctly predicted negative observations to total observations. (D) The harmonic mean of true positives and true negatives.
ANSWER: (B) The ratio of correctly predicted positive observations to total predicted positive observations.

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In a fire alarm system, if the model predicts "Fire Present" when there is actually no fire, this is classified as:
OPTIONS: (A) True Positive (TP) (B) True Negative (TN) (C) False Positive (FP) (D) False Negative (FN)
ANSWER: (C) False Positive (FP)

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: In the context of autonomous vehicle safety systems, which type of error would be most critical to minimize?
OPTIONS: (A) False Positive (detecting danger when there isn't any) (B) False Negative (failing to detect actual danger) (C) True Positive (detecting danger correctly) (D) True Negative (correctly identifying that there is no danger)
ANSWER: (B) False Negative (failing to detect actual danger)

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: An AI model was tested with 1000 test samples. If True Positive (TP)= 200, True Negative (TN)= 600, False Positive (FP)= 100, False Negative (FN)= 100, how many total predictions were correct?
OPTIONS: (A) 300 (B) 600 (C) 800 (D) 900
ANSWER: (C) 800

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Statement 1: Overfitting occurs when a model memorizes the training data rather than learning patterns. Statement 2: Using the same data for training and evaluation helps the model give accurate results.
OPTIONS: (A) Both statements are correct. (B) Both statements are incorrect. (C) Statement 1 is correct but statement 2 is incorrect. (D) Statement 1 is incorrect but statement 2 is correct.
ANSWER: (C) Statement 1 is correct but statement 2 is incorrect.

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: What is the primary need for evaluating an AI model's performance in the AI Model Development process?
OPTIONS: (A) To increase the complexity of the model. (B) To visualize the data. (C) To assess how well the chosen model will work in future. (D) To reduce the amount of data used for training.

ANSWER: (C) To assess how well the chosen model will work in future.

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Statement 1: To evaluate a models' performance, we need either precision or recall. Statement 2: When the value of both Precision and Recall is 1, the F1 score is 0.
OPTIONS: (A) Both statement 1 and statement 2 are correct. (B) Both statement 1 and statement 2 are incorrect. (C) Statement 1 is correct, but statement 2 is incorrect. (D) Statement 1 is incorrect, but statement 2 is correct.
ANSWER: (C) Statement 1 is correct, but statement 2 is incorrect.

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Suppose you are developing an AI model to detect fraudulent financial transaction risk. Describe False Positives and False Negatives in this context.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: False Positives are legitimate transactions incorrectly classified as fraudulent (Reality: NO, Prediction: YES). False Negatives are fraudulent transactions not detected by the model (Reality: YES, Prediction: NO).

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Explain Train-test split technique with respect to machine learning algorithm.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Train-test split is a technique for evaluating machine learning algorithm performance by dividing the dataset into training and testing subsets, typically in 80-20% or 70-30% ratio.

YEAR: 2025
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: B

ORIGINAL_MARKS: 4
QUESTION: An AI model has been developed to test specimens of blood/urine/cough etc. to diagnose ailments (diabetes/liver infection etc.). The model was tested on a data-set of about 630 tests and the resulting confusion matrix is provided. (A) How many total cases are True Negative? (B) Calculate Precision, Recall and F1 Score.
OPTIONS: This is a numerical question based on confusion matrix. Not converted to MCQ.
ANSWER: True Negative = 410. Precision = 110/170 ≈ 0.647, Recall = 110/160 ≈ 0.6875, F1 Score ≈ 0.667.

YEAR: 2026
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In supervised learning, what is the purpose of the testing dataset?
OPTIONS: (A) To train the model. (B) To evaluate the model's accuracy. (C) To create new features. (D) To label the data.
ANSWER: (B) To evaluate the model's accuracy.

YEAR: 2026
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In a fire alarm system, if the model predicts "Fire Present" when there is actually no fire, this is classified as:
OPTIONS: (A) True Positive (TP) (B) True Negative (TN) (C) False Positive (FP) (D) False Negative (FN)
ANSWER: (C) False Positive (FP)

YEAR: 2026
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the context of autonomous vehicle safety systems, which type of error would be most critical to minimize?
OPTIONS: (A) False Positive (detecting danger when there isn't any) (B) False Negative (failing to detect actual danger) (C) True Positive (detecting danger correctly) (D) True Negative (correctly identifying that there is no danger)
ANSWER: (B) False Negative (failing to detect actual danger)

YEAR: 2026
CHAPTER: 3 - Evaluating Models
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: An AI model was tested with 1000 test samples. If True Positive (TP)= 200, True Negative (TN)= 600, False Positive (FP)= 100, False Negative (FN)= 100, how many total predictions were correct?
OPTIONS: (A) 300 (B) 600 (C) 800 (D) 900
ANSWER: (C) 800

==
CHAPTER: 4 - Statistical Data
==

YEAR: 2023
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: What is Dimensionality Reduction?
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Dimensionality reduction is the process of reducing the number of random variables under consideration, by obtaining a set of principal variables.

YEAR: 2024
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is the correct expansion of CSV?
OPTIONS: (A) Comma and Space Value (B) Comma Separated Values (C) Control Separated Values (D) Character Separated Values
ANSWER: (B) Comma Separated Values

YEAR: 2024
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: Which AI domain would be most suitable for developing a price comparison website?
OPTIONS: (A) Computer Vision (B) Natural Language Processing (C) Statistical Data (D) Robotics
ANSWER: (C) Statistical Data

YEAR: 2024
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Differentiate between Machine Learning (ML) and Deep Learning (DL).
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: ML is a subset of AI that uses algorithms to learn from data; DL is a subset of ML that uses neural networks with many layers to learn from large amounts of data.

YEAR: 2025
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is the correct expansion of CSV?
OPTIONS: (A) Comma and Space Value (B) Comma Separated Values (C) Control Separated Values (D) Character Separated Values
ANSWER: (B) Comma Separated Values

YEAR: 2025
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: It is a domain-specific language that is designed for managing data held in different kinds of DBMS. It is particularly useful in handling structured data. Which computer language is this?
OPTIONS: (A) SQL (B) CSV (C) Spreadsheet (D) TXT
ANSWER: (A) SQL

YEAR: 2025

CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which AI domain would be most suitable for developing a price comparison website?
OPTIONS: (A) Computer Vision (B) Natural Language Processing (C) Statistical Data (D) Robotics
ANSWER: (C) Statistical Data

YEAR: 2025
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: One of the applications of Data Science is Airline Route Planning. List any two tasks that airline companies can do using Data Science.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (1) Predict flight delays. (2) Determine optimal routes (direct or with halts).

YEAR: 2026
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is the correct expansion of CSV?
OPTIONS: (A) Comma and Space Value (B) Comma Separated Values (C) Control Separated Values (D) Character Separated Values
ANSWER: (B) Comma Separated Values

YEAR: 2026
CHAPTER: 4 - Statistical Data
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which AI domain would be most suitable for developing a price comparison website?
OPTIONS: (A) Computer Vision (B) Natural Language Processing (C) Statistical Data (D) Robotics
ANSWER: (C) Statistical Data

==
CHAPTER: 5 - Computer Vision
==

YEAR: 2022
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 1
QUESTION: Name any two currently popular virtual assistants.
OPTIONS: (A) Siri and Alexa (B) Google Maps and YouTube (C) Windows and Linux (D) Photoshop and Illustrator
ANSWER: (A) Siri and Alexa

YEAR: 2023
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Email filters, spam filters, smart assistants are the examples of:
OPTIONS: (A) Pocket Assistants (B) CV (C) NLP (D) Evaluation
ANSWER: (C) NLP

YEAR: 2023
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Smart Assistants such as Alexa, Siri are the examples of:
OPTIONS: (A) Natural Language Processing (B) Data Science (C) Machine Learning (D) Computer Vision
ANSWER: (A) Natural Language Processing

YEAR: 2023
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Face lock feature of a smartphone is an example of computer vision. Briefly discuss this feature.
OPTIONS: This is a subjective question. Not converted to MCQ.

ANSWER: Face lock uses computer vision to detect and recognise a user's face by analysing facial features, and unlock the device when a match is found.

YEAR: 2024
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Anita and Surjit are creating an AI application that will classify different types of fruits. The Computer Vision task that will identify the type of fruit and assign a label to it is called:
OPTIONS: (A) Segmentation (B) Classification (C) Classification + Localization (D) Object Detection
ANSWER: (B) Classification

YEAR: 2024
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In computer vision which of the following tasks is used for multiple objects?
OPTIONS: (A) Classification (B) Classification + Localisation (C) Instance Segmentation (D) Localisation
ANSWER: (C) Instance Segmentation

YEAR: 2024
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): The term used to refer to the number of pixels in an image is resolution. Reason (R): Resolution in an image denotes the total number of pixels it contains, usually represented as height × width.
OPTIONS: (A) Both (A) and (R) are true and (R) is the correct explanation for (A). (B) Both (A) and (R) are true and (R) is not the correct explanation for (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.
ANSWER: (A) Both (A) and (R) are true and (R) is the correct explanation for (A).

YEAR: 2024
CHAPTER: 5 - Computer Vision

ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The concept of ________ is used to apply face filters on various social media platforms.
OPTIONS: (A) NLP (B) Computer Vision (C) Data Science (D) Block chain Technology
ANSWER: (B) Computer Vision

YEAR: 2024
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Differentiate between grayscale and RGB images.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Grayscale images have one channel with pixel values 0-255 (shades of gray). RGB images have three channels (Red, Green, Blue) each with 0-255, combining to produce colour.

YEAR: 2024
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: How do computers store RGB images?
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: RGB images are stored as three separate channels (R, G, B). Each pixel has three values (0-255) representing intensity of each colour, combining to give the final colour.

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Anita and Surjit are creating an AI application that will classify different types of fruits. The Computer Vision task that will identify the type of fruit and assign a label to it is called:
OPTIONS: (A) Segmentation (B) Classification (C) Classification + Localization (D) Object Detection
ANSWER: (B) Classification

YEAR: 2025

CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: What does the term "image processing" refer to in Computer Vision?
OPTIONS: (A) Editing videos (B) Extracting meaningful information from images (C) Playing audio files (D) Compiling codes
ANSWER: (B) Extracting meaningful information from images

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In Computer Vision, which of the following tasks is used for single object?
OPTIONS: (A) Object Detection (B) Classification + Localization (C) Instance Segmentation (D) Non-Localization
ANSWER: (B) Classification + Localization

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Raghav can turn on and off any appliance remotely using an internet enabled device. This is an example of:
OPTIONS: (A) Artificial Intelligence (AI) (B) Internet of Things (IoT) (C) Computer Vision (CV) (D) Deep Learning (DL)
ANSWER: (B) Internet of Things (IoT)

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Statement 1: Various search engines and e-commerce portals now have a new feature called image-based search using computer vision. Statement 2: Image-based search helps in finding items, people and places by giving their sounds to the system.
OPTIONS: (A) Both Statement 1 and Statement 2 are correct. (B) Both Statement 1 and Statement 2 are incorrect. (C) Statement 1 is correct but Statement 2 is incorrect. (D) Statement 2 is correct but Statement 1 is incorrect.
ANSWER: (C) Statement 1 is correct but Statement 2 is incorrect.

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In computer vision which of the following tasks is used for multiple objects?
OPTIONS: (A) Classification (B) Classification + Localisation (C) Instance Segmentation (D) Localisation
ANSWER: (C) Instance Segmentation

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): The term used to refer to the number of pixels in an image is resolution. Reason (R): Resolution in an image denotes the total number of pixels it contains, usually represented as height × width.
OPTIONS: (A) Both (A) and (R) are true and (R) is the correct explanation for (A). (B) Both (A) and (R) are true and (R) is not the correct explanation for (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.
ANSWER: (A) Both (A) and (R) are true and (R) is the correct explanation for (A).

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The concept of ________ is used to apply face filters on various social media platforms.
OPTIONS: (A) NLP (B) Computer Vision (C) Data Science (D) Block chain Technology
ANSWER: (B) Computer Vision

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: What is the range of possible pixel values in a byte image format?

OPTIONS: (A) 0 to 100 (B) 0 to 255 (C) 1 to 256 (D) -128 to 127
ANSWER: (B) 0 to 255

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: How do computers store RGB images?
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: All colored images are made up of three primary colors Red, Green, and Blue. Every RGB image is stored in the form of three different channels called the R Channel, G Channel, and B Channel. Each plane separately has many pixels with each value varying from 0-255.

YEAR: 2025
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Differentiate between Computer Vision (CV) and Natural Language Processing (NLP).
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Computer Vision deals with visual information (images/videos), while NLP deals with text and speech. CV extracts information from pixels; NLP from words and sentences.

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Anita and Surjit are creating an AI application that will classify different types of fruits. The Computer Vision task that will identify the type of fruit and assign a label to it is called:
OPTIONS: (A) Segmentation (B) Classification (C) Classification + Localization (D) Object Detection
ANSWER: (B) Classification

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: What does the term "image processing" refer to in Computer Vision?
OPTIONS: (A) Editing videos (B) Extracting meaningful information from images (C) Playing audio files (D) Compiling codes
ANSWER: (B) Extracting meaningful information from images

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In Computer Vision, which of the following tasks is used for single object?
OPTIONS: (A) Object Detection (B) Classification + Localization (C) Instance Segmentation (D) Non-Localization
ANSWER: (B) Classification + Localization

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Raghav can turn on and off any appliance remotely using an internet enabled device. This is an example of:
OPTIONS: (A) Artificial Intelligence (AI) (B) Internet of Things (IoT) (C) Computer Vision (CV) (D) Deep Learning (DL)
ANSWER: (B) Internet of Things (IoT)

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Statement 1: Various search engines and e-commerce portals now have a new feature called image-based search using computer vision. Statement 2: Image-based search helps in finding items, people and places by giving their sounds to the system.
OPTIONS: (A) Both Statement 1 and Statement 2 are correct. (B) Both Statement 1 and Statement 2 are incorrect. (C) Statement 1 is correct but Statement 2 is incorrect. (D) Statement 2 is correct but Statement 1 is incorrect.
ANSWER: (C) Statement 1 is correct but Statement 2 is incorrect.

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In computer vision which of the following tasks is used for multiple objects?
OPTIONS: (A) Classification (B) Classification + Localisation (C) Instance Segmentation (D) Localisation
ANSWER: (C) Instance Segmentation

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): The term used to refer to the number of pixels in an image is resolution. Reason (R): Resolution in an image denotes the total number of pixels it contains, usually represented as height × width.
OPTIONS: (A) Both (A) and (R) are true and (R) is the correct explanation for (A). (B) Both (A) and (R) are true and (R) is not the correct explanation for (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.
ANSWER: (A) Both (A) and (R) are true and (R) is the correct explanation for (A).

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The concept of ________ is used to apply face filters on various social media platforms.
OPTIONS: (A) NLP (B) Computer Vision (C) Data Science (D) Block chain Technology
ANSWER: (B) Computer Vision

YEAR: 2026
CHAPTER: 5 - Computer Vision
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: What is the range of possible pixel values in a byte image format?
OPTIONS: (A) 0 to 100 (B) 0 to 255 (C) 1 to 256 (D) -128 to 127
ANSWER: (B) 0 to 255

==
CHAPTER: 6 - Natural Language Processing (NLP)
==

YEAR: 2022
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Differentiate between Script-bot and Smart-bot.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Script-bots follow predefined scripts, are rigid and require coding; Smart-bots are flexible, AI-powered, work on bigger databases, and can understand natural language.

YEAR: 2022
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 4
QUESTION: Consider the text of following documents: Document 1: Sahil likes to play cricket, Document 2: Sajal likes cricket too, Document 3: Sajal also likes to play basketball. Apply all the four steps of Bag of words model of NLP on the above given documents and generate the output.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Step 1: Text Normalization (lowercase, remove punctuation). Step 2: Create vocabulary of unique words. Step 3: Create document vectors (counts). Step 4: Form document-term matrix.

YEAR: 2022
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 4
QUESTION: With reference to NLP, explain the following terms in detail with the help of suitable example: Term frequency, Inverse Document Frequency.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Term Frequency (TF) measures how frequently a term appears in a document. Inverse Document Frequency (IDF) measures how important a term is by down-weighting common terms. Example: In a document corpus, "the" has high TF but low IDF, while "NLP" has high IDF.

YEAR: 2023
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: For the whole corpus is divided into sentences. Each sentence is taken as a different data so now the whole corpus gets reduced to sentences.
OPTIONS: (A) Text Regulation (B) Sentence Segmentation (C) Tokenisation (D) Stemming
ANSWER: (B) Sentence Segmentation

YEAR: 2023
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a feature of document classification?
OPTIONS: (A) Helps in classifying the type and genre of a document. (B) Helps in creating a document. (C) Helps to display important information of a corpus. (D) Helps in including the necessary words in the text body.
ANSWER: (A) Helps in classifying the type and genre of a document.

YEAR: 2023
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: With reference to data processing, expand the term TFIDF. Also give any two applications of TFIDF.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: TF-IDF stands for Term Frequency-Inverse Document Frequency. Applications: information retrieval, text summarization, and document clustering.

YEAR: 2024
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: This real life application of NLP is used to provide an overview of a news item or blog post, while avoiding redundancy from multiple sources and maximising the diversity of content obtained. Which is this application?
OPTIONS: (A) Chatbot (B) Virtual Assistant (C) Sentiment Analysis (D) Automatic Summarisation
ANSWER: (D) Automatic Summarisation

YEAR: 2024
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following words represent an example of a lemma resulting from lemmatisation for "caring" in context to Natural Language Processing (NLP)?
OPTIONS: (A) Care (B) Cared (C) Cares (D) Car
ANSWER: (A) Care

YEAR: 2024
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In spam email detection, which of the following will be considered as "False Negative"?
OPTIONS: (A) When a legitimate email is accurately identified as not spam. (B) When a spam email is mistakenly identified as legitimate. (C) When an email is accurately recognised as spam. (D) When an email is inaccurately labelled as important.
ANSWER: (B) When a spam email is mistakenly identified as legitimate.

YEAR: 2024
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which domain of AI is used for interacting with virtual assistants such as Siri and Alexa?
OPTIONS: (A) Machine Learning (ML) (B) Computer Vision (CV) (C) Natural Language Processing (NLP) (D) Technical Vision (TV)
ANSWER: (C) Natural Language Processing (NLP)

YEAR: 2024
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: What are the primary differences between Script-bots and Smart-bots?
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Script-bots are rule-based and require coding; Smart-bots are AI-powered, flexible, and can learn from interactions.

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A corpus contains 4 documents in which the words such as 'an, is, the' were appearing frequently. Identify the term that is used for such words.
OPTIONS: (A) Stop word (B) Rare word (C) Missing word (D) Removable word
ANSWER: (A) Stop word

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which application of NLP helps to provide an overview of a news item or blog post? It also avoids redundancy from multiple sources and maximises the diversity of content obtained.
OPTIONS: (A) Virtual Assistants (B) Sentiment Analysis (C) Text Classification (D) Automatic Summarization
ANSWER: (D) Automatic Summarization

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: On seeing her son's result, Pooja's face turned red with anger. The word "red" demonstrates which characteristic of natural language?
OPTIONS: (A) Redundancy (B) Context-dependent meaning (C) Grammatical structure (D) Temporal change

ANSWER: (B) Context-dependent meaning

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which NLP application helps in converting natural speech into text in real time?
OPTIONS: (A) Keyword Extraction tool (B) Translation of books from English to Hindi language (C) Auto generated captions on YouTube (D) Classifying raw text into pre-defined groups
ANSWER: (C) Auto generated captions on YouTube

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A company wants to analyze customer reviews to understand satisfaction levels. Which NLP application would be most suitable?
OPTIONS: (A) Text classification (B) Sentiment analysis (C) Keyword extraction (D) Language translation
ANSWER: (B) Sentiment analysis

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which type of chat-bot has a wide functionality, is flexible and powerful, and works on bigger databases directly?
OPTIONS: (A) Script bot (B) Smart bot (C) Traditional bot (D) Rule-based bot
ANSWER: (B) Smart bot

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following words represent an example of a lemma resulting from lemmatisation for "caring" in context to Natural Language Processing (NLP)?

OPTIONS: (A) Care (B) Cared (C) Cares (D) Car
ANSWER: (A) Care

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the context of NLP, which of the following words represents a stem resulting from stemming for "Studies"?
OPTIONS: (A) Study (B) Stud (C) Studi (D) Studied
ANSWER: (C) Studi

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which domain of AI is used for interacting with virtual assistants such as Siri and Alexa?
OPTIONS: (A) Machine Learning (ML) (B) Computer Vision (CV) (C) Natural Language Processing (NLP) (D) Technical Vision (TV)
ANSWER: (C) Natural Language Processing (NLP)

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Bag of Words is a model which helps in extracting features out of the text which can be helpful in machine learning algorithms. This belongs to:
OPTIONS: (A) Data Science (DS) (B) Virtual Reality (VR) (C) Natural Language Processing (NLP) (D) Computer Vision (CV)
ANSWER: (C) Natural Language Processing (NLP)

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2

QUESTION: Differentiate between Script-bot and Smart-bot.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Script-bots are rule-based and require coding; Smart-bots are AI-powered, flexible, and can learn from interactions.

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: What is Tokenization? Count how many tokens are present in the following statement: "I find that the harder I work, the more luck I seem to have."
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Tokenization is the process of breaking text into smaller units called tokens. The given sentence has 15 tokens.

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 4
QUESTION: Consider the text of following documents: Document 1: Sahil likes to play cricket, Document 2: Sajal likes cricket too, Document 3: Sajal also likes to play basketball. Apply all the four steps of Bag of words model of NLP on the above given documents and generate the output.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (Steps as described in earlier answer)

YEAR: 2025
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: With reference to NLP, explain the following terms in detail with the help of suitable example: Term frequency, Inverse Document Frequency.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (Definition and example as earlier)

YEAR: 2025

CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 4
QUESTION: Consider the following documents: Document 1: NLP is a domain of AI. Document 2: NLP stands for Natural Language Processing. Implement all the four steps of Bag of Words (BoW) model to create a document vector table.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Steps: normalize text, create vocabulary, count frequencies, form document-term matrix.

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A corpus contains 4 documents in which the words such as 'an, is, the' were appearing frequently. Identify the term that is used for such words.
OPTIONS: (A) Stop word (B) Rare word (C) Missing word (D) Removable word
ANSWER: (A) Stop word

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which application of NLP helps to provide an overview of a news item or blog post? It also avoids redundancy from multiple sources and maximises the diversity of content obtained.
OPTIONS: (A) Virtual Assistants (B) Sentiment Analysis (C) Text Classification (D) Automatic Summarization
ANSWER: (D) Automatic Summarization

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: On seeing her son's result, Pooja's face turned red with anger. The word "red" demonstrates which characteristic of natural language?
OPTIONS: (A) Redundancy (B) Context-dependent meaning (C) Grammatical structure (D) Temporal change

ANSWER: (B) Context-dependent meaning

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which NLP application helps in converting natural speech into text in real time?
OPTIONS: (A) Keyword Extraction tool (B) Translation of books from English to Hindi language (C) Auto generated captions on YouTube (D) Classifying raw text into pre-defined groups
ANSWER: (C) Auto generated captions on YouTube

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A company wants to analyze customer reviews to understand satisfaction levels. Which NLP application would be most suitable?
OPTIONS: (A) Text classification (B) Sentiment analysis (C) Keyword extraction (D) Language translation
ANSWER: (B) Sentiment analysis

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which type of chat-bot has a wide functionality, is flexible and powerful, and works on bigger databases directly?
OPTIONS: (A) Script bot (B) Smart bot (C) Traditional bot (D) Rule-based bot
ANSWER: (B) Smart bot

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following words represent an example of a lemma resulting from lemmatisation for "caring" in context to Natural Language Processing (NLP)?

OPTIONS: (A) Care (B) Cared (C) Cares (D) Car
ANSWER: (A) Care

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the context of NLP, which of the following words represents a stem resulting from stemming for "Studies"?
OPTIONS: (A) Study (B) Stud (C) Studi (D) Studied
ANSWER: (C) Studi

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which domain of AI is used for interacting with virtual assistants such as Siri and Alexa?
OPTIONS: (A) Machine Learning (ML) (B) Computer Vision (CV) (C) Natural Language Processing (NLP) (D) Technical Vision (TV)
ANSWER: (C) Natural Language Processing (NLP)

YEAR: 2026
CHAPTER: 6 - Natural Language Processing
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: How is Stemming different from Lemmatization? Explain how the word "wolves" would be processed by stemming and lemmatization.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Stemming produces "wolv" (non-meaningful), lemmatization produces "wolf" (meaningful). Stemming is faster but less accurate.

==
CHAPTER: Employability Skills - Communication Skills
==

YEAR: 2022
CHAPTER: Employability Skills - Communication Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following types of communication takes place when the number of people is small enough to communicate with each other effectively?
OPTIONS: (A) Interpersonal communication (B) Public communication (C) Intrapersonal communication (D) Small group communication
ANSWER: (D) Small group communication

YEAR: 2023
CHAPTER: Employability Skills - Communication Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Identify the imperative sentence:
OPTIONS: (A) Shut the front door. (B) She is talented artist. (C) Are you feeling better? (D) You were amazing!
ANSWER: (A) Shut the front door.

YEAR: 2024
CHAPTER: Employability Skills - Communication Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following types of communication takes place when one individual addresses a large gathering?
OPTIONS: (A) Written communication (B) Public communication (C) Small group communication (D) Interpersonal communication
ANSWER: (B) Public communication

YEAR: 2025
CHAPTER: Employability Skills - Communication Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Identify the imperative sentence:
OPTIONS: (A) Shut the front door. (B) She is talented artist. (C) Are you feeling better? (D) You were amazing!
ANSWER: (A) Shut the front door.

YEAR: 2025
CHAPTER: Employability Skills - Communication Skills
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Write the 7 C's of communication.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Clear, Concise, Concrete, Correct, Coherent, Complete, Courteous.

YEAR: 2025
CHAPTER: Employability Skills - Communication Skills
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: If you are a team leader of a team of 20 people in an organization, mention any two methods that you will use for effective communication with your team members.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (1) Face-to-face meetings; (2) Business meetings or emails.

YEAR: 2026
CHAPTER: Employability Skills - Communication Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Identify the imperative sentence:
OPTIONS: (A) Shut the front door. (B) She is talented artist. (C) Are you feeling better? (D) You were amazing!
ANSWER: (A) Shut the front door.

==
CHAPTER: Employability Skills - Self-management Skills
==

YEAR: 2022
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is not a step to build self-motivation?

OPTIONS: (A) Focusing on your goal. (B) Planning to achieve your goal. (C) Being indisciplined. (D) Finding out your strength.
ANSWER: (C) Being indisciplined.

YEAR: 2023
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a quality of successful entrepreneurs?
OPTIONS: (A) Hard working (B) Resistance to change (C) Lazy (D) Less-confident
ANSWER: (A) Hard working

YEAR: 2024
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: SMART method can be used to set goals to make you successful in your career and personal life. What does 'A' in SMART stand for?
OPTIONS: (A) Abrupt (B) Accountable (C) Achievable (D) Admirable
ANSWER: (C) Achievable

YEAR: 2024
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is not a key element of self-management skills?
OPTIONS: (A) Prioritising your work (B) Not taking feedback (C) Goal setting (D) Staying updated about new practices
ANSWER: (B) Not taking feedback

YEAR: 2025
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is not a step to build self-motivation?

OPTIONS: (A) Focusing on your goal. (B) Planning to achieve your goal. (C) Being indisciplined. (D) Finding out your strength.
ANSWER: (C) Being indisciplined.

YEAR: 2025
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following does not help in stress management?
OPTIONS: (A) Healthy food (B) Sound sleep (C) Yoga asanas (D) Negative thoughts
ANSWER: (D) Negative thoughts

YEAR: 2025
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Breaking down big goals into smaller parts will make the goal:
OPTIONS: (A) specific (B) measurable (C) achievable (D) realistic
ANSWER: (C) achievable

YEAR: 2025
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Explain the importance of following a healthy lifestyle in effectively dealing with stress. Write any one common factor that causes stress among the children nowadays.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: A healthy lifestyle (diet, exercise, sleep) helps manage stress. A common factor: academic pressure.

YEAR: 2026
CHAPTER: Employability Skills - Self-management Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following does not help in stress management?
OPTIONS: (A) Healthy food (B) Sound sleep (C) Yoga asanas (D) Negative thoughts

ANSWER: (D) Negative thoughts

==
CHAPTER: Employability Skills - ICT Skills
==

YEAR: 2022
CHAPTER: Employability Skills - ICT Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: GUI stands for:
OPTIONS: (A) Graphical User Interaction (B) Graphical User Interface (C) Graphical Upper Interface (D) None of these
ANSWER: (B) Graphical User Interface

YEAR: 2023
CHAPTER: Employability Skills - ICT Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Spam refers to:
OPTIONS: (A) Unnecessary images (B) Temporary files (C) Junk mails (D) Music files
ANSWER: (C) Junk mails

YEAR: 2024
CHAPTER: Employability Skills - ICT Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The most important software in any computer is the _______. This is the software that starts working as soon as we switch on a computer.
OPTIONS: (A) Web Browsers (B) Operating System (C) Office Software (D) Designing Software
ANSWER: (B) Operating System

YEAR: 2025
CHAPTER: Employability Skills - ICT Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: What should a strong password consist of?
OPTIONS: (A) Only letters (B) Numbers and special characters (C) Name of a person (D) Letters, numbers and special characters
ANSWER: (D) Letters, numbers and special characters

YEAR: 2025
CHAPTER: Employability Skills - ICT Skills
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Enlist any two measures that an individual should follow to take care of his/her digital devices.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (1) Use updated antivirus. (2) Regularly update software and back up data.

YEAR: 2026
CHAPTER: Employability Skills - ICT Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: What should a strong password consist of?
OPTIONS: (A) Only letters (B) Numbers and special characters (C) Name of a person (D) Letters, numbers and special characters
ANSWER: (D) Letters, numbers and special characters

==
CHAPTER: Employability Skills - Entrepreneurial Skills
==

YEAR: 2022
CHAPTER: Employability Skills - Entrepreneurial Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is not a task of an entrepreneur?
OPTIONS: (A) Sharing of wealth (B) Preferably using foreign materials (C) Fulfilling customer needs (D) Helping society
ANSWER: (B) Preferably using foreign materials

YEAR: 2023
CHAPTER: Employability Skills - Entrepreneurial Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a function of an entrepreneur?
OPTIONS: (A) Following the traditional method of business (B) Innovation (C) Keeping all the profit to himself/herself (D) Avoid taking decisions
ANSWER: (B) Innovation

YEAR: 2024
CHAPTER: Employability Skills - Entrepreneurial Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Reema has started her own restaurant. She keeps on trying new ideas to make different dishes for her customers. As an entrepreneur, Reema is:
OPTIONS: (A) Impatient (B) Creative (C) Under-confident (D) Lazy
ANSWER: (B) Creative

YEAR: 2025
CHAPTER: Employability Skills - Entrepreneurial Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A misconception about an entrepreneur is: Entrepreneurs are ______, not ______.
OPTIONS: (A) born, made (B) rich, poor (C) unique, common (D) big businessmen, small businessmen
ANSWER: (A) born, made

YEAR: 2025
CHAPTER: Employability Skills - Entrepreneurial Skills
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Write any two tasks that entrepreneurs do when they run their business.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: (1) Decision making (2) Managing business operations.

YEAR: 2026

CHAPTER: Employability Skills - Entrepreneurial Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A misconception about an entrepreneur is: Entrepreneurs are ______, not ______.
OPTIONS: (A) born, made (B) rich, poor (C) unique, common (D) big businessmen, small businessmen
ANSWER: (A) born, made

==
CHAPTER: Employability Skills - Green Skills
==

YEAR: 2022
CHAPTER: Employability Skills - Green Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: How food is one of the major problems related to sustainable development? Discuss briefly.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Food problem: wastage, unequal distribution, and unsustainable agricultural practices lead to food insecurity and environmental degradation.

YEAR: 2023
CHAPTER: Employability Skills - Green Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): Sustainable agriculture is environment friendly. Reason (R): It prevents use of chemical fertilizers to protect soil.
OPTIONS: (A) Both (A) and (R) are correct and (R) is the correct explanation of (A). (B) Both (A) and (R) are correct but (R) is not the correct explanation of (A). (C) (A) is correct but (R) is not correct. (D) (A) is not correct but (R) is correct.
ANSWER: (A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

YEAR: 2024
CHAPTER: Employability Skills - Green Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Organic farming technique is an example of a green skill that is essential for sustainable agriculture. Reason (R): Organic farming technique prioritise environment friendly and sustainable practices such as using natural fertilisers, avoiding synthetic pesticides and promoting soil health.
OPTIONS: (A) Both (A) and (R) are true and (R) is the correct explanation for (A). (B) Both (A) and (R) are true and (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.
ANSWER: (A) Both (A) and (R) are true and (R) is the correct explanation for (A).

YEAR: 2025
CHAPTER: Employability Skills - Green Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Choose the option which is NOT a Sustainable Development Goal according to United Nations.
OPTIONS: (A) Population (B) No poverty (C) Quality education (D) Reduced Inequalities
ANSWER: (A) Population

YEAR: 2025
CHAPTER: Employability Skills - Green Skills
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Define the term 'Sustainable Development'.
OPTIONS: This is a subjective question. Not converted to MCQ.
ANSWER: Sustainable Development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

YEAR: 2026
CHAPTER: Employability Skills - Green Skills
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): Organic farming technique is an example of a green skill that is essential for sustainable agriculture. Reason (R): Organic farming technique prioritise environment friendly and sustainable practices such as using natural fertilisers, avoiding synthetic pesticides and promoting soil health.
OPTIONS: (A) Both (A) and (R) are true and (R) is the correct explanation for (A). (B) Both (A) and (R) are true and (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

ANSWER: (A) Both (A) and (R) are true and (R) is the correct explanation for (A).
